Forensic Investigation of Slack: Data Persistence After User-Initiated Deletion

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May 2026

Abstract

This white paper presents the findings of a capstone research project conducted at Bridgewater State University examining forensic artifacts retained by Slack across mobile and desktop environments. The investigation explores how data persistence varies after user-initiated deletion, with a focus on differences across device type, user role, and channel visibility.

A controlled environment comprising one administrator account and two standard user accounts was deployed across an iPhone 12 on iOS 26.3 and a Lenovo ThinkPad T480 running Windows 10. Realistic Slack activity was simulated using a free trial of Slack's paid tier, including message creation and deletion, file sharing across public and private channels, and huddle recordings. Forensic images were acquired and analyzed using FTK Imager 8.2, supplemented by SlackDump (an open-source, API-based extraction tool) and a Slack-native administrator data export.

Findings demonstrate that Slack deletion is not as final as officially documented. Significant artifacts persisted on Windows endpoints and through API-based extraction. The iPhone iTunes backup yielded no recoverable Slack artifacts, revealing a notable gap between mobile and desktop forensic recovery. No single tool was sufficient on its own; FTK Imager and SlackDump are complementary methods that each capture distinct categories of evidence.

1. INTRODUCTION

Slack is a workplace messaging and collaboration platform that has become one of the most widely used communication tools in professional environments. Launched in 2013 and acquired by Salesforce in 2021, Slack organizes team communication into channels dedicated to specific topics, projects, or departments, and in doing so handles significant quantities of sensitive operational and personal data.

Its widespread adoption across corporate, academic, and government environments makes Slack an increasingly relevant target in digital forensics investigations, internal audits, and legal discovery proceedings. A critical assumption embedded in Slack's official documentation is that user-initiated deletion is permanent and irreversible. This research challenges that assumption directly, providing forensic investigators with a grounded assessment of what data can realistically be recovered from Slack deployments following deletion.

An additional forensic consideration involves Slack's Electron-based desktop application, which encrypts local SQLite databases via SQLCipher. This is not documented anywhere by Slack; it functions as an incidental barrier that standard forensic tools cannot parse without the encryption key. This paper documents both the capabilities and limitations of current forensic methods when applied to Slack environments and establishes the complementary roles of disk imaging and API-based extraction in a complete investigative workflow.

2. RESEARCH QUESTIONS

This project was structured around three core investigative questions:

- What user data artifacts persist on mobile and desktop devices after user-initiated deletion in Slack?
- Can channel content be partially reconstructed through endpoint comparison across an administrator account and two standard user accounts?
- How does recoverability vary across attachment types (images, documents, spreadsheets) by device type and extraction method?

These questions guided all aspects of the investigation, from environment configuration and simulated activity through acquisition, analysis, and interpretation of results.

3. METHODOLOGY

3.1 Controlled Environment Setup

The investigation used a controlled Slack workspace built on a free trial of Slack's paid tier, which provided access to the full feature set including administrative export capabilities. Three accounts were provisioned across two device types:

- iPhone 12, iOS 26.3— Bob Stone (standard user)
- Lenovo ThinkPad T480, Windows 10 — Alice Stone (standard user)
- Propeller Windows VM — Chris Stevens (administrator)

Bob and Alice served as the primary accounts, handling the bulk of messaging and file sharing activity. Chris's role in the channels was deliberately limited; he sent one message in the private channel and one in the public channel. His account was included to test whether administrator-level access and exports would yield forensically different results compared to standard user extractions.

All three accounts participated in both a private channel ("capstone") and a public channel ("new-channel"). Dummy data was transmitted across both channels before all messages and files were fully deleted by each user, simulating a scenario in which a subject attempts to remove evidence of prior activity.

3.2 Simulated Activity

The following data types were transmitted within the private channel prior to deletion:

- Microsoft Excel spreadsheets
- Microsoft Word documents
- Text messages with attachments
- Text messages without attachments
- Images (PNG, JPEG)
- A Slack Huddle (recorded meeting)

Activity also occurred in the public channel. Chris sent "Hi Team!" on March 2; Bob replied "Hey Chris so excited to join" the same day, with a follow-up exchange on March 16. In the private channel, Chris sent "Hello Team" on March 16 and Bob replied "Welcome" in the huddle thread. After all activity was complete, every message and file was fully deleted by each respective user account prior to acquisition.

3.3 Acquisition Methods

Following complete deletion, three acquisition methods were employed:

- FTK Imager 8.2 — Forensic disk images were acquired for both Windows devices. An iTunes logical backup was used for the iPhone. Forensic image integrity was verified using both MD5 and SHA1 hashes, which matched between stored and calculated values, confirming no tampering or corruption occurred during acquisition.
- SlackDump — An open-source, API-based extraction tool executed against all three accounts. SlackDump exports channel content and workspace metadata as JSON, operating independently of local device storage. It was introduced mid-project as a pivot after iPhone imaging challenges consumed a significant portion of available project time.
- Slack-Native Export — An administrator-initiated data export through the Chris Stevens account, requested for use as a comparison baseline against SlackDump output.

3.4 Verification Challenges

A critical verification challenge arose during Windows analysis: distinguishing artifacts that originated from Slack's local channel cache versus files present in the user's Downloads folder from the initial test data setup. Some findings initially appeared to be Slack cache artifacts but were confirmed to have originated from the pre-acquisition download process. Careful file path analysis was applied throughout to avoid false positives and maintain evidentiary integrity. This is a real-world forensic concern that investigators should anticipate in any Slack-related case.

4. RESULTS

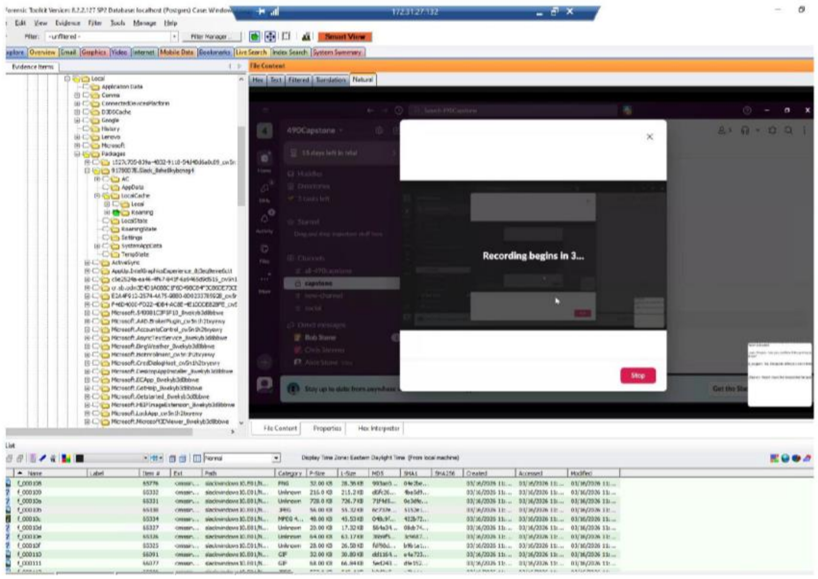
4.1 Windows Forensic Analysis (FTK Imager — Alice Stone)

The Windows analysis of Alice's device was the most productive component of the investigation. All significant artifacts were recovered from the following Slack cache directory:

```
Users\Alice\AppData\Local\Packages\91750D7E.Slack_8she8kybcnzg4\LocalCache\Roaming\Slack\Cache\Cache_Data\
```

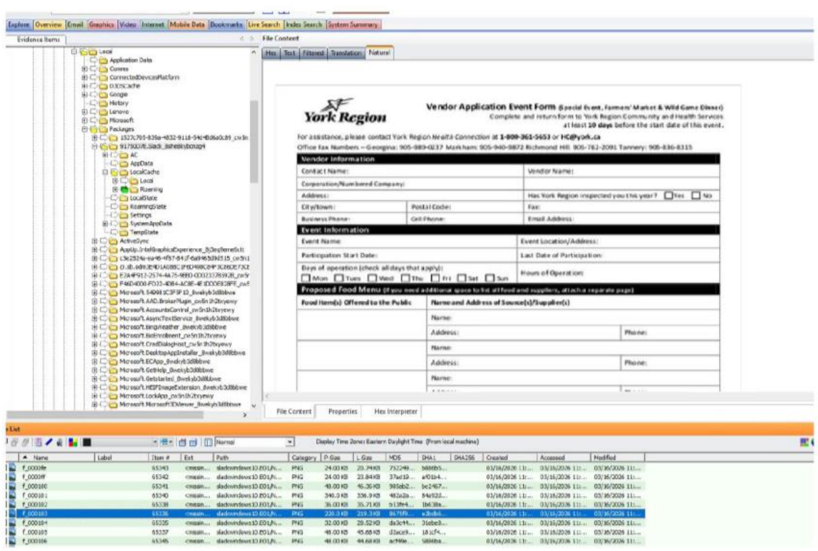
Everything stored in this cache was in plaintext; no encryption was applied to cached files. The following artifacts were recovered:

- Cached file attachments in plaintext: images (PNG, JPEG), Excel spreadsheets, and Word documents. Notably, these included files sent by other users (Bob and Chris), not only files Alice sent herself. This finding demonstrates that a single endpoint can yield forensic evidence about the activity of multiple channel participants.
- A cached screenshot of the recorded Huddle, capturing the full Slack interface including the sidebar, channel list, user list, and a "Recording begins in 3..." countdown. This single artifact provided significant contextual information about workspace state at the time of the recording.
- Slack cookies (.slack.com) recovered from a Chrome Cookies SQLite database, providing session metadata with creation and expiration timestamps.
- The root-state.json file, containing Alice's user ID, workspace domain (490capstone-workspace), workspace URL, and workspace metadata.



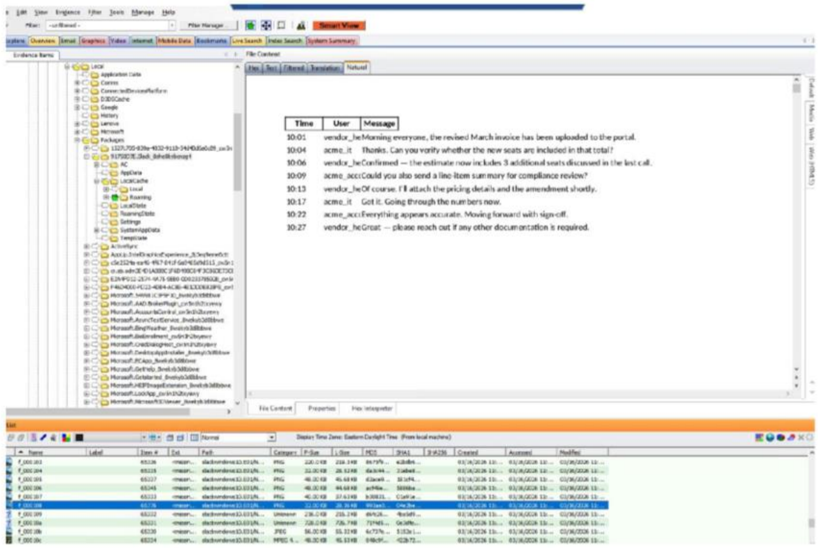
slackwindows10.E01/NONAME
[NTFS]/[root]/Users/Alice/AppData/Local/Packages/91750D7E.Slack_8she8kybcnzg4/LocalCache/Roaming/Slack/Cache/Cache_Data/f_00010b

Figure 1. FTK Imager view of a cached Huddle screenshot (f_00010b) recovered from Alice's Slack cache. The full Slack interface is visible, including the sidebar, channel list, and a "Recording begins in 3..." countdown.

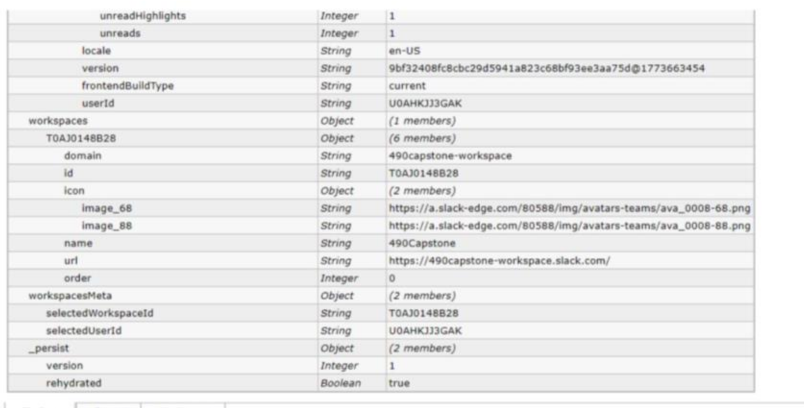


slackwindows10.E01/NONAME
[NTFS]/[root]/Users/Alice/AppData/Local/Packages/91750D7E.Slack_8she8kybcnzg4/LocalCache/Roaming/Slack/Cache/Cache_Data/f_000103

Figure 2. FTK Imager view of the cached Vendor Application Event Form PDF (f_000103), a file shared in the channel by another user and recovered in plaintext from Alice's local Slack cache.



slackwindows10.E01/NONAME [NTFS]/[root]/Users/Alice/AppData/Local/Packages/91750D7E.Slack_8she8kybcnzg4/LocalCache/Roaming/Slack/Cache/Cache_Data/f_000108



unreadHighlights	Integer	1
unreads	Integer	1
locale	String	en-US
version	String	9bf32408fc8cbc29d5941a823c68bf93ee3aa75d@1773663454
frontendBuildType	String	current
userId	String	U0AHKJJ3GAK
workspaces	Object	(1 members)
T0AJ0148B28	Object	(6 members)
domain	String	490capstone-workspace
id	String	T0AJ0148B28
icon	Object	(2 members)
image_68	String	https://a.slack-edge.com/80588/img/avatars-teams/ava_0008-68.png
image_88	String	https://a.slack-edge.com/80588/img/avatars-teams/ava_0008-88.png
name	String	490Capstone
url	String	https://490capstone-workspace.slack.com/
order	Integer	0
workspacesMeta	Object	(2 members)
selectedWorkspaceId	String	T0AJ0148B28
selectedUserId	String	U0AHKJJ3GAK
_persist	Object	(2 members)
version	Integer	1
rehydrated	Boolean	true

File Content | Properties | Hex Interpreter

Figure 3. FTK Imager view of recovered Excel spreadsheet content (f_000108) showing vendor coordination messages in tabular format, alongside the parsed root-state.json output revealing Alice's user ID, workspace domain, and workspace URL.

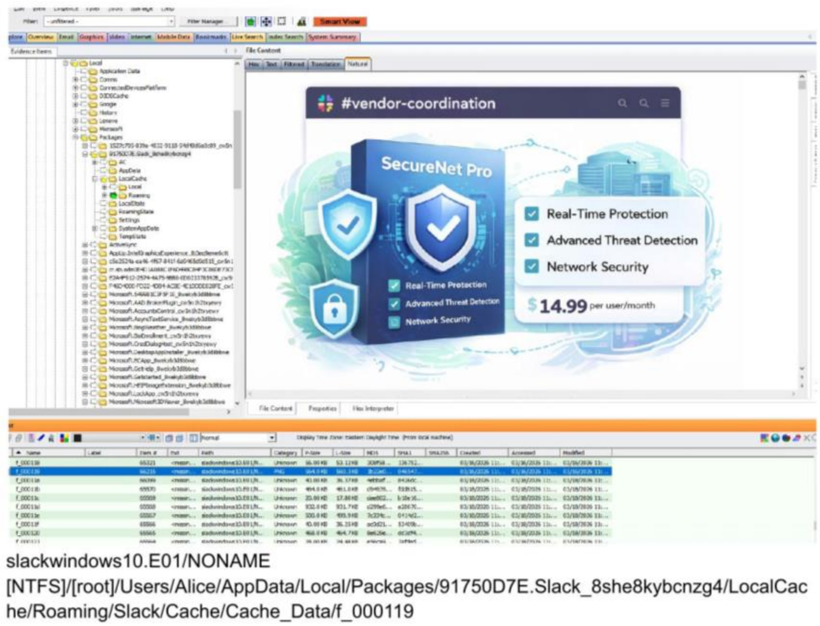


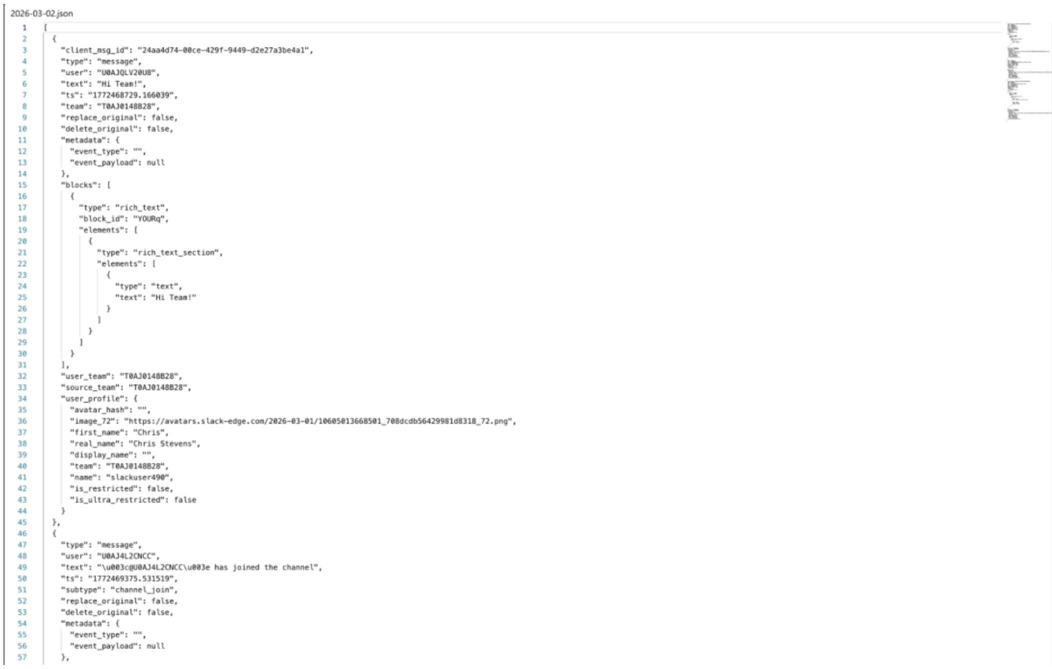
Figure 4. FTK Imager view of a cached image file (f_000119) recovered from the Slack cache directory, showing a SecureNet Pro advertisement originally shared in the channel.

Text messages were not recoverable via FTK. Slack's local SQLite databases are encrypted using SQLCipher 4 with the following parameters: page size 4,096; 256,000 KDF iterations; SHA-512 HMAC. This configuration is undocumented in Slack's public documentation and cannot be parsed by standard forensic tools without the encryption key. It is an incidental barrier built into the Electron-based desktop application.

The administrator's Windows image (Chris Stevens) was acquired but not fully analyzed. Time consumed by iPhone imaging challenges prevented completion of that portion of the analysis within the project timeline.

4.2 SlackDump API-Based Extraction

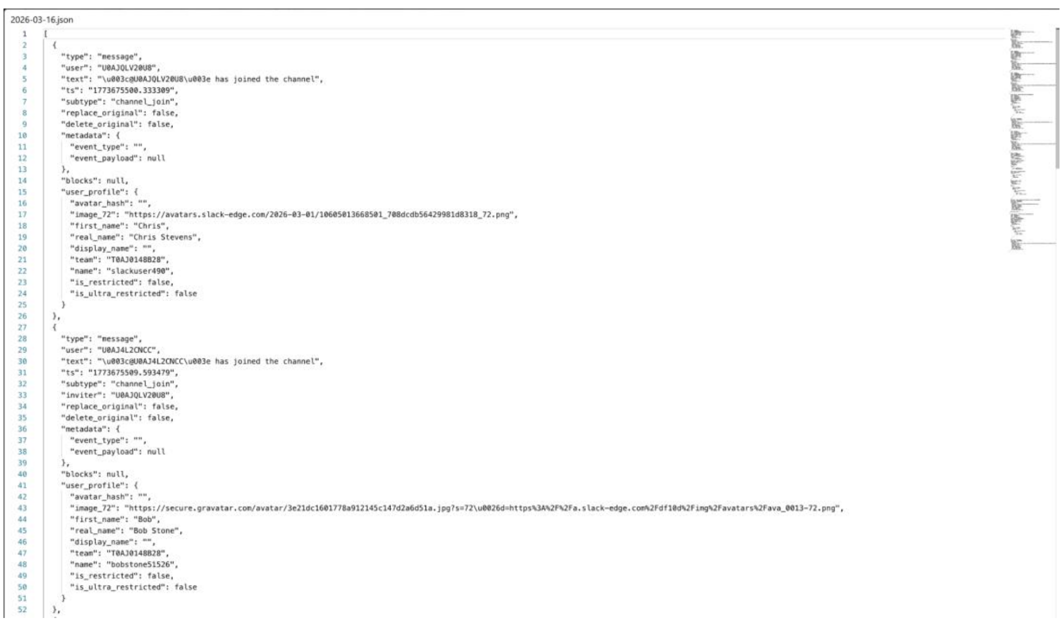
SlackDump was executed against all three accounts. Results were consistent regardless of user role or activity level.



2026-03-02.json

```
[
  {
    "client_msg_id": "24aa4d74-00ce-429f-9449-d2e27a3be4a1",
    "type": "message",
    "user": "U0AJQLV20U8",
    "text": "Hi Team!",
    "ts": "1772468729.166039",
    "team": "T0AJ0148B28",
    "replace_original": false,
    "delete_original": false,
    "metadata": {
      "event_type": "",
      "event_payload": null
    },
    "blocks": [
      {
        "type": "rich_text",
        "block_id": "YOURq",
        "elements": [
          {
            "type": "rich_text_section",
            "elements": [
              {
                "type": "text",
                "text": "Hi Team!"
              }
            ]
          }
        ]
      }
    ],
    "user_team": "T0AJ0148B28",
    "source_team": "T0AJ0148B28",
    "user_profile": {
      "avatar_hash": "",
      "image_72": "https://avatars.slack-edge.com/2026-03-01/10605013668501_708dcdb56429981d8318_72.png",
      "first_name": "Chris",
      "real_name": "Chris Stevens",
      "display_name": "",
      "team": "T0AJ0148B28",
      "name": "slackuser490",
      "is_restricted": false,
      "is_ultra_restricted": false
    }
  },
  {
    "type": "message",
    "user": "U0AJ4L2CNCC",
    "text": "\u003c@U0AJ4L2CNCC\u003e has joined the channel",
    "ts": "1772469375.531519",
    "subtype": "channel_join",
    "replace_original": false,
    "delete_original": false,
    "metadata": {
      "event_type": "",
      "event_payload": null
    },
```


Figure 5. SlackDump JSON output from the public channel (2026-03-02.json) showing recovered messages including Chris Stevens' "Hi Team!" and Bob Stone's reply, both fully intact after deletion.



2026-03-16.json

```
[
  {
    "type": "message",
    "user": "U0AJQLV20U8",
    "text": "\u003c@U0AJQLV20U8\u003e has joined the channel",
    "ts": "1773675500.333309",
    "subtype": "channel_join",
    "replace_original": false,
    "delete_original": false,
    "metadata": {
      "event_type": "",
      "event_payload": null
    },
    "blocks": null,
    "user_profile": {
      "avatar_hash": "",
      "image_72": "https://avatars.slack-edge.com/2026-03-01/10605013668501_708dcdb56429981d8318_72.png",
      "first_name": "Chris",
      "real_name": "Chris Stevens",
      "display_name": "",
      "team": "T0AJ0148B28",
      "name": "slackuser490",
      "is_restricted": false,
      "is_ultra_restricted": false
    }
  },
  {
    "type": "message",
    "user": "U0AJ4L2CNCC",
    "text": "\u003c@U0AJ4L2CNCC\u003e has joined the channel",
    "ts": "1773675509.593479",
    "subtype": "channel_join",
    "inviter": "U0AJQLV20U8",
    "replace_original": false,
    "delete_original": false,
    "metadata": {
      "event_type": "",
      "event_payload": null
    },
    "blocks": null,
    "user_profile": {
      "avatar_hash": "",
      "image_72": "https://secure.gravatar.com/avatar/3e21dc1601778a912145c147d2a6d51a.jpg?s=72\u0026d=https%3A%2F%2Fa.slack-edge.com%2Fdf10d%2Fimg%2Favatars%2Fava_0013-72.png",
      "first_name": "Bob",
      "real_name": "Bob Stone",
      "display_name": "",
      "team": "T0AJ0148B28",
      "name": "bobstone51526",
      "is_restricted": false,
      "is_ultra_restricted": false
    }
  },
```


Figure 6. SlackDump JSON output from the private channel (2026-03-16.json) showing channel join events with inviter metadata confirming that Chris Stevens (U0AJQLV20U8) invited both Bob and Alice.

2026-03-16.json


```json
    }
  ],
  "user_team": "T0AJ0148B28",
  "source_team": "T0AJ0148B28",
  "user_profile": {
    "avatar_hash": "",
    "image_72": "https://avatars.slack-edge.com/2026-03-01/10605013668501_708dcdb56429981d8318_72.png",
    "first_name": "Chris",
    "real_name": "Chris Stevens",
    "display_name": "",
    "team": "T0AJ0148B28",
    "name": "slackuser490",
    "is_restricted": false,
    "is_ultra_restricted": false
  }
},
{
  "type": "message",
  "user": "U0AJ4L2CNCC",
  "text": "set the channel topic: Project ",
  "ts": "1773676235.334489",
  "subtype": "channel_topic",
  "topic": "Project ",
  "replace_original": false,
  "delete_original": false,
  "metadata": {
    "event_type": "",
    "event_payload": null
  },
  "blocks": null,
  "user_profile": {
    "avatar_hash": "",
    "image_72": "https://secure.gravatar.com/avatar/3e21dc1601778a912145c147d2a6d51a.jpg?s=72\u0026d=https%3A%2F%2Fa.slack-edge.com%2Fdf10d%2Fimg%2Favatars%2Fava_0013-72.png",
    "first_name": "Bob",
    "real_name": "Bob Stone",
    "display_name": "",
    "team": "T0AJ0148B28",
    "name": "bobstone51526",
    "is_restricted": false,
    "is_ultra_restricted": false
  }
},
{
  "type": "message",
  "channel": "C0ALU85DHGE",
  "user": "USLACKBOT",
  "ts": "1773676257.396729",
  "thread_ts": "1773676257.396729",
  "edited": {
    "user": "USLACKBOT",
    "ts": "1773676676.000000"
  },
  "last_read": "1773676257.396729",
  "subscribed": true,
  "subtype": "huddle_thread",
  "reply_count": 1,
  "reply_users": [
    "U0AJ4L2CNCC"
  ],
  "replies": [
    {
      "user": "U0AJ4L2CNCC",
      "ts": "1773676286.283739"
    }
```


Figure 7. SlackDump JSON output showing Bob Stone's "set the channel topic: Project" event and the huddle thread metadata, including the Slackbot-posted call permalink and reply structure.

users.json


```json
},
{
  "id": "U0AJ4L2CNCC",
  "team_id": "T0AJ0148B28",
  "name": "bobstone51526",
  "deleted": false,
  "color": "d58247",
  "real_name": "Bob Stone",
  "tz": "America/New_York",
  "tz_label": "Eastern Daylight Time",
  "tz_offset": -14400,
  "profile": {
    "first_name": "Bob",
    "last_name": "Stone",
    "real_name": "Bob Stone",
    "real_name_normalized": "Bob Stone",
    "display_name": "",
    "display_name_normalized": "",
    "avatar_hash": "g3e21dc16017",
    "email": "bobstone51526@gmail.com",
    "image_24": "https://secure.gravatar.com/avatar/3e21dc1601778a912145c147d2a6d51a.jpg?s=24\u0026d=https%3A%2F%2Fa.slack-edge.com%2Fdf10d%2Fimg%2Favatars%2Fava_0013-24.png",
    "image_32": "https://secure.gravatar.com/avatar/3e21dc1601778a912145c147d2a6d51a.jpg?s=32\u0026d=https%3A%2F%2Fa.slack-edge.com%2Fdf10d%2Fimg%2Favatars%2Fava_0013-32.png",
    "image_48": "https://secure.gravatar.com/avatar/3e21dc1601778a912145c147d2a6d51a.jpg?s=48\u0026d=https%3A%2F%2Fa.slack-edge.com%2Fdf10d%2Fimg%2Favatars%2Fava_0013-48.png",
    "image_72": "https://secure.gravatar.com/avatar/3e21dc1601778a912145c147d2a6d51a.jpg?s=72\u0026d=https%3A%2F%2Fa.slack-edge.com%2Fdf10d%2Fimg%2Favatars%2Fava_0013-72.png",
    "image_192": "https://secure.gravatar.com/avatar/3e21dc1601778a912145c147d2a6d51a.jpg?s=192\u0026d=https%3A%2F%2Fa.slack-edge.com%2Fdf10d%2Fimg%2Favatars%2Fava_0013-192.png",
    "image_512": "https://secure.gravatar.com/avatar/3e21dc1601778a912145c147d2a6d51a.jpg?s=512\u0026d=https%3A%2F%2Fa.slack-edge.com%2Fdf10d%2Fimg%2Favatars%2Fava_0013-512.png",
    "huddle_state": "default_unset",
    "team": "T0AJ0148B28",
    "fields": []
  },
  "is_bot": false,
  "is_admin": false,
  "is_owner": false,
  "is_primary_owner": false,
  "is_restricted": false,
  "is_ultra_restricted": false,
  "is_stranger": false,
  "is_app_user": false,
  "is_invited_user": false,
  "is_email_confirmed": true,
  "has_2fa": false,
  "two_factor_type": null,
  "has_files": false,
  "presence": "",
  "locale": "en-US",
  "updated": 1774289193,
  "who_can_share_contact_card": "EVERYONE",
  "enterprise_user": {
    "id": "",
    "enterprise_id": "",
    "enterprise_name": "",
    "is_admin": false,
    "is_owner": false,
    "is_primary_owner": false,
    "teams": null
  }
```


Figure 8. SlackDump users.json output showing a recovered user profile for Bob Stone, including real name, email address, avatar URLs, timezone, and admin/owner status flags.

channels.json


```
[
  {
    "id": "C0AHWCS8AEN",
    "created": 1772420995,
    "is_open": false,
    "last_read": "1772639606.768689",
    "is_group": false,
    "is_shared": false,
    "is_im": false,
    "is_ext_shared": false,
    "is_org_shared": false,
    "is_global_shared": false,
    "is_pending_ext_shared": false,
    "is_private": false,
    "is_read_only": false,
    "is_mpim": false,
    "unlinked": 0,
    "name_normalized": "all-490capstone",
    "num_members": 3,
    "priority": 0,
    "user": "",
    "shared_team_ids": [
      "T0AJ0148B28"
    ],
```


Figure 9. SlackDump channels.json output showing recovered channel metadata for the "all-490capstone" channel, including privacy settings, member count, and shared team IDs.

Content Viewer [dms.json]

Hex | Text | Filtered | Translation | Natural


```
[
  {
    "id": "D0AHXKELFB7",
    "created": 1772468678,
    "members": [
      "U0AJQLV20U8",
      "U0AHKJJ3GAK"
    ]
  },
  {
    "id": "D0AJ12E9VRQ",
    "created": 1772468678,
    "members": [
      "U0AHKJJ3GAK",
      "U0AHKJJ3GAK"
    ]
  },
  {
    "id": "D0AJ4L3N9LL",
    "created": 1772468679,
    "members": [
      "U0AJ4L2CNCC",
      "U0AHKJJ3GAK"
    ]
  },
  {
    "id": "D0AJDVAQL1F",
    "created": 1772468678,
    "members": [
      "USLACKBOT",
      "U0AHKJJ3GAK"
    ]
  }
]
```


Figure 10. SlackDump dms.json output showing recovered direct message conversation records, listing DM channel IDs, creation timestamps, and member pairs including Slackbot.

Recovered from the public channel ("new-channel"):

- Chris Stevens: "Hi Team!" (March 2)
- Bob Stone: "Hey Chris so excited to join" (March 2)
- Bob Stone: "Haha all good" (March 16)
- All user join events with timestamps

Recovered from the private channel ("capstone"):

- Chris Stevens: "Hello Team" (March 16)
- Bob Stone: "Welcome" (reply in huddle thread, March 16)
- Huddle metadata including a permalink to the call and Chris's raised-hands emoji reaction
- Bob setting the channel topic to "Project"
- All user join events, including inviter metadata confirming that Chris invited both Bob and Alice

Recovered from workspace metadata:

- Full user profiles: real names, email addresses, usernames, time zones, avatar URLs, and admin/owner status. These confirmed Chris as admin, owner, and primary owner of the workspace.
- Channel structure for all four channels, including privacy settings, creation dates, creators, member lists, and previous channel names. The general channel was identified as having been renamed from "all-new-workspace" to "all-490capstone."
- DM conversation records identifying which users had direct message conversations with each other, including Slackbot.

Any message sent with an attachment was not recoverable. Once deleted, both the attached file and the accompanying message text disappeared entirely; the attachment did not vanish in isolation but removed the message record with it.

The Slack-native administrator export through Chris's account produced results identical to SlackDump. Administrator privileges provided no additional forensic data beyond what standard user-level API extraction yielded.

4.3 iPhone Forensic Analysis

iPhone imaging was the most time-consuming challenge of the project, consuming a significant portion of the total available project time. The iTunes logical backup ultimately yielded no significant Slack artifacts. No messages, cached files, images, or metadata were recovered from Bob Stone's iPhone 12.

A more advanced extraction method, such as a full filesystem extraction, may produce different results; however, this was outside the scope of the current project given the time constraints encountered. The iPhone result establishes a clear gap between mobile and desktop forensic recovery for Slack under standard acquisition methods.

4.4 Recoverability Summary

Data Type	FTK Imager (Windows — Alice)	SlackDump / Slack Export
Messages (no attachment)	Not Recovered	**Recovered**
Messages (with attachment)	Not Recovered	Not Recovered
Images (PNG, JPEG)	**Recovered**	Not Recovered
Excel Spreadsheets	**Recovered**	Not Recovered
Word Documents	**Recovered**	Not Recovered
Huddle Screenshot	**Recovered**	Metadata Only
Cookies / Session Data	**Recovered**	N/A
User Profiles & Metadata	N/A	**Recovered**
Channel Structure	N/A	**Recovered**
iPhone Artifacts	Not Recovered	Not Recovered

Table 1. Recoverability of data types by acquisition method across device environments.

5. KEY FINDINGS

- **Slack deletion is not as permanent as officially documented. Significant artifacts persisted on Windows endpoints and were recoverable via API-based extraction after complete user-initiated deletion.**
- No single tool recovered everything. FTK Imager and SlackDump are complementary: FTK captures cached files and session data; SlackDump captures message text and workspace metadata.
- A single endpoint can yield evidence about multiple accounts. Alice's Windows image contained cached files originating from Bob's and Chris's activity, not only Alice's own.
- Administrator privileges provided no forensic advantage. SlackDump and the Slack-native admin export produced identical results across all three accounts regardless of user role.

- The iPhone yielded nothing under iTunes logical backup, establishing a significant gap between mobile and desktop forensic recovery for Slack.
- SQLCipher 4 encryption on Slack's local SQLite databases is an undocumented forensic barrier. Standard tools cannot overcome it without the encryption key; investigators should anticipate and document this obstacle.
- Messages sent with attachments represent a genuine data loss scenario. Once deleted, neither the file nor the accompanying message text was recoverable by any method tested.
- Adaptability is essential in forensic casework. The pivot from iPhone analysis to SlackDump, driven by real-world time constraints, produced some of the project's most meaningful findings.

6. SIGNIFICANCE AND IMPLICATIONS

As a platform widely used for sensitive business communication, Slack is increasingly relevant to digital forensics, internal audits, and legal proceedings. The findings of this research carry practical implications across several stakeholder groups.

For Digital Forensic Investigators

Investigators examining Slack-related cases should prioritize Windows endpoint analysis through cache directory examination and complement it with API-based extraction using tools like SlackDump. These methods are not redundant; they capture different categories of evidence and together provide the most comprehensive coverage. SQLCipher encryption should be anticipated on any Slack installation and documented as an identified barrier in case reports.

For Legal and Compliance Professionals

Slack's claim that user-initiated deletion is permanent is contradicted by empirical evidence. Forensic recovery of deleted content is possible under certain conditions, which carries direct implications for e-discovery, spoliation arguments, and data retention policy enforcement. Organizations and legal teams relying on Slack's deletion documentation as a basis for data management decisions should revisit those assumptions.

For Organizations Using Slack

Organizations should be aware that local device caches may retain data after deletion from the Slack server. Endpoint security policies, including cache clearing and device management controls, should account for Slack's caching behavior. The administrator export function provides no special advantage over user-level API access for recovering deleted content.

Privacy and Ethical Considerations

The ability to recover deleted content raises significant privacy concerns. Direct messages carry heightened privacy expectations in real investigations; a warrant or legal authorization would be required to access DM history in an actual case. All evidence in this study was obtained through authorized methods within a controlled academic environment using simulated accounts and data. No real personal data was used or exposed. Organizations implementing monitoring through admin exports must maintain clear, disclosed policies to avoid misuse.

7. LIMITATIONS

- The administrator's Windows image was acquired but not fully analyzed due to time constraints caused by iPhone imaging challenges.
- iPhone analysis was limited to an iTunes logical backup. Advanced extraction methods such as full filesystem extraction or jailbreak-based acquisition were outside the project scope.
- Only one private channel and one public channel were tested. Behavior may differ in more complex workspace configurations with greater message volumes or varied permission structures.
- Distinguishing Slack cache artifacts from files in the user's local Downloads folder required careful verification; some initial findings were false positives that were identified and excluded.
- The study used a free trial of Slack's paid tier. Results may differ across subscription tiers with different server-side data retention policies.

8. FUTURE WORK

- Complete analysis of the administrator's Windows forensic image to determine whether admin-level local caching differs from standard user caching behavior.
- Explore SQLCipher decryption approaches, including memory forensics methods to extract the encryption key from a running Slack process.

- Test advanced iOS extraction methods such as jailbreak-based full filesystem extraction or third-party mobile forensic platforms to determine whether Slack artifacts are present but inaccessible via logical backup.
- Investigate whether Slack version updates or update cadence affects cache retention and the structure of recovered artifacts.
- Examine differences in artifact recovery between public and private channels in greater depth, including more complex workspace configurations.

9. CONCLUSION

This capstone project provides verifiable evidence that Slack's user-initiated deletion process does not fully remove data from local device endpoints. Using FTK Imager, SlackDump, and a Slack-native administrator export, the research team recovered a meaningful range of artifacts including cached file attachments, session cookies, message text, and workspace metadata after complete deletion across all accounts.

The findings reinforce a foundational principle of digital forensics: deletion from an application's interface does not guarantee deletion from the underlying storage systems. For Slack specifically, Windows desktop endpoints retain a rich cache of channel activity that persists post-deletion, while iPhone devices present a significantly higher forensic barrier under standard acquisition methods.

For digital forensic investigators, the complementary use of disk imaging tools and API-based extraction offers the most comprehensive coverage. Neither approach alone is sufficient. Investigators should be prepared to encounter SQLCipher encryption as an undocumented obstacle and document it accordingly in case reports. The adaptability demonstrated in this project, pivoting from iPhone imaging to API extraction under real-world time constraints, reflects the practical reality of forensic casework and the importance of methodological flexibility.

APPENDIX: TOOLS AND ENVIRONMENT

Tool / Resource	Description / Role in Study
FTK Imager 8.2	Forensic disk imaging and analysis for Windows devices; iTunes backup acquisition of the iPhone
SlackDump	Open-source, API-based extraction tool for Slack workspaces; produced JSON output of message content and workspace metadata
Slack (Paid Trial)	Platform under investigation; free trial of paid tier used to enable full feature set including administrative export capabilities

SQLCipher 4	Encryption layer on Slack's local SQLite databases; page size 4096, 256,000 KDF iterations, SHA-512 HMAC
iPhone 12 (iOS 26.3)	Mobile device — Bob Stone (standard user)
Lenovo ThinkPad T480 (Windows 10)	Desktop device — Alice Stone (standard user)
Propeller Windows VM	Virtual machine — Chris Stevens (administrator); limited channel activity
iTunes Logical Backup	iPhone acquisition method; yielded no Slack artifacts

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